SAURABH RAWAT

Computer Science & Engineering Student • Graphic Era Hill University

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EDUCATION

Undergraduate Record

Graphic Era Hill University — B.Tech in Computer Science & Engineering **Academic Record: 9+ CGPA in 1st & 2nd Year** `Currently Pursuing (Aug 2022 – Present)`

ISC Board (Class XII): Intermediate (89%)

ICSE Board (Class X): High School (94%)

TECHNICAL SKILLS

Programming Languages:	C, C++, Java, Python, JavaScript, TypeScript, SQL
Web Development:	React.js, Next.js, Node.js, Express.js, HTML5, CSS3, REST APIs, Tailwind CSS, Socket.IO, Vite
Databases & ORM:	PostgreSQL, SQLite, Supabase, Neon, Prisma ORM, Firebase Realtime DB
AI/ML & Computer Vision:	Machine Learning, Computer Vision, YOLOv8, InsightFace, DeepSORT, OpenCV, PaddleOCR, TensorFlow.js, FastAPI
IoT & Systems:	Sensor Integration, Embedded C++, IoT Prototyping (Arduino Uno), Serial Telemetry, Geospatial Systems
Developer Tools & DevOps:	Git, GitHub, VS Code, Vercel, Docker, Docker Compose, Postman
CS Fundamentals & Quality:	Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, API & Unit Testing, Debugging

TECHNICAL EXPERIENCE & PROJECTS

Kaarobarie POS Billing Software

— Multi-Tenant Restaurant POS & Management SaaS

- **Architected** a multi-tenant restaurant management SaaS with multi-franchise tenancy, visual table occupancy management, automated billing, and live kitchen order ticket (KOT) workflows using React, TypeScript, Node.js, Express, Prisma ORM, and PostgreSQL (Supabase), achieving sub-15ms p95 API response times and zero-leakage tenant isolation.
- **Engineered** 7-tier hierarchical Role-Based Access Control (RBAC) and real-time Socket.IO bi-directional event routing for kitchen display synchronization, automated payroll calculations, and ESC/POS thermal receipt printing (58mm/80mm), eliminating order coordination latency between floor staff and kitchen stations.

Railway Gangmen Safety System

— Safety-Focused Embedded Railway Monitoring & Early Warning System

- **Developed** an embedded safety-monitoring and early warning system to protect railway track maintenance personnel (gangmen) from oncoming train collision hazards using C++, Arduino Uno, proximity/vibration sensors, and RF transceivers, achieving real-time approach detection and instantaneous local audio-visual buzzer alarms.
- **Implemented** a low-latency multi-node telemetry pipeline with threshold-based sensor calibration and fail-safe signal broadcasting via C++ and Python serial communication listeners, significantly improving track-worker situational awareness across hazardous blind-curve operational zones.

AI-Powered Missing Children Finder

— Distributed Computer Vision & Trajectory Analysis Search Grid

- **Engineered** an AI-powered surveillance search pipeline for identifying and tracking missing children across asynchronous CCTV footage streams using Python, FastAPI, Ultralytics YOLOv8, InsightFace (ArcFace embeddings), DeepSORT, and OpenCV, extracting 512-dimensional facial vector embeddings for high-accuracy identity re-identification.
- **Integrated** automated vehicle number-plate recognition (ANPR) using PaddleOCR alongside camera-network trajectory graph traversal within a Dockerized microservice and Next.js Leaflet map HUD, automating multi-camera movement forecasting and reducing manual surveillance auditing times during critical emergency response windows.

IoT-Based Landslide Detection System

— Geological Instability Monitoring & Early Warning Network

- **Developed** an IoT geological monitoring and early warning system to detect slope displacement and imminent landslides in vulnerable terrains using Embedded C++, MPU-6050 accelerometer/gyroscope sensors, soil moisture probes, and ADC modules, sampling angle tilt and soil saturation deltas with continuous anomaly threshold filtering.
- **Implemented** edge-to-cloud serial data transmission and automated emergency alerting logic using C++, Python, and REST endpoints, delivering instant multi-level hazard alerts (Advisory, Warning, Critical) to downstream monitoring consoles upon breach of geological slope stability thresholds.

AI-Based Pothole Detection System

— Edge Computer Vision & Geo-Spatial Road Defect Alert Platform

- **Engineered** a real-time road surface defect and pothole detection system utilizing client-side edge computer vision on live video streams with TensorFlow.js, a custom in-browser CNN model (fjs_model), HTML5 Canvas, and Web APIs, classifying road surface damage directly on the client without cloud compute costs.
- **Integrated** automated GPS geolocation coordinate stamping and real-time cloud dispatch into Firebase Realtime Database with transactional deduplication, logging sub-meter geographical coordinates and hazard severity indices to prioritize municipal road repair workflows.

MANAGEMENT & LEADERSHIP EXPERIENCE

IEEE Student Branch, GEHU

`Member & Organising Committee`

Key Initiatives: TechnIEEEks 2025, SAARTHI 2025 (24-Hr National Hackathon), TechnIEEEks 2026

- **Coordinated** technical operations, workshop logistics, and participant onboarding for flagship initiatives including TechnIEEEks 2025, SAARTHI 2025 (24-hour national hackathon), and TechnIEEEks 2026.
- **Demonstrated** structured event coordination and cross-functional teamwork, facilitating transparent communication between faculty advisors, external technical mentors, and student attendees across multi-day sessions.

Technology Business Incubator (TBI), GEU

`Member & Organising Committee`

Key Initiatives: Graphethon 2.0 (48-Hr), Graphethon 3.0 (72-Hr National Hackathons)

- **Managed** on-ground event operations and participant registration workflows for 500+ attendees during the 48-hour Graphethon 2.0 and 72-hour Graphethon 3.0 flagship national hackathons.
- **Applied** proactive team leadership and operational problem-solving to coordinate venue logistics, mentor schedules, and crisis communications, ensuring uninterrupted 24/7 hackathon sprint execution.

HACKATHON ACHIEVEMENTS

- **SELECTED** **National-Level Hackathons Selection** — Team selected for national-level hackathons across competitive development sprints, including MINDCRAFT 2K26 (Top-40 Selection).
- **SEMIFINALIST** **Economic Times X GenAI Hackathon** — Semifinalist in nationwide generative AI engineering competition.
- **FINALIST** **PRAXIS 2.0 (Google Developer Groups)** — Finalist in national technical hackathon building scalable software solutions.
- **FINALIST** **Neural Nexus AI/ML Challenge** — Finalist at Grafest 2026 machine learning & vision challenge.